

Shing Kai Lai

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EDUCATION

Rice University

M.Sc. Mechanical Engineering (Expected Graduation Dec 2026).

Houston, Texas

Aug 2025 – Present

Hong Kong University of Science and Technology (HKUST)

B.Eng. Aerospace Engineering; Extended Major in Artificial Intelligence.

Hong Kong

Sep 2021 – Jun 2025

SKILLS

Programming Languages: Python, C/C++, MATLAB, Simulink, LabVIEW, Bash/Linux Shell, Arduino

Software: SolidWorks, COMSOL, Ansys Fluent/Mechanical, NVIDIA Omniverse, Isaac Sim

Industrial Tools: CNC machining, robotic arm programming, Bambu 3D printer, Formlabs SLA Printer, sheet-metal and glass fabrication tools, general manufacturing and laboratory tools

WORK EXPERIENCE

Research Scientist

Dec 2025 – Present

Rice University

Houston, Texas

- Integrated LabVIEW and Python toolpath generation with Aerotech three-axis motion control and custom magnetic-field actuation to print functional filaments with programmable magnetic-pole orientation.
- Designed the DIW platform in SolidWorks and integrated the dispensing nozzle, magnetic-field unit, and Aerotech motion system into a functional customized printer.
- Optimized DIW ink formulations through literature-guided studies and printability testing of extrusion behavior, filament geometry, and layer stability.

Research Scientist

Aug 2025 – Dec 2025

Rice University

Houston, Texas

- Developed an electrically triggered gastric-resident drug-release device with a multidisciplinary team, using voltage-driven metal dissolution for controlled drug release.
- Designed the drug reservoir and electrochemical testing platform in SolidWorks; fabricated multi-material prototypes using a Bambu 3D printer.
- Deposited a $\sim 2\ \mu\text{m}$ gold-film anode via e-beam evaporation and modeled voltage-dependent dissolution via Tafel kinetics to control release timing.
- Designed a mold-casting system in SolidWorks with drug-storage chambers and microneedle structures; cast using Dragon Skin silicone.

Research Assistant

May 2024 – Sep 2024

Embry-Riddle Aeronautical University

Daytona Beach, Florida

- Programmed a Rainbow Robotics RB16-900 six-axis robotic arm and a three-axis dispensing robot for vibration-assisted coaxial printing using MATLAB-generated toolpaths.
- Designed a multi-outlet coaxial dispensing component in SolidWorks, validated iterations using an SLA printer, and manufactured the final component through CNC machining.
- Enabled fabrication of a 400 x 210 x 60 mm granular-composite structure; evaluated mechanical performance using an MTS universal testing machine and characterized internal structure and epoxy infiltration using Bruker X-ray micro-CT.
- Prepared epoxy-based printable ink using a Thinky ARV-310P planetary mixer, Onilab centrifuge, and pneumatic dispenser.
- Co-authored "Vibrational 3D Printing of Granular Materials Toward In-Space Construction," submitted to Advanced Materials.

PROJECT EXPERIENCE

Robotics Learning and Teleoperation Data Pipeline

Sep 2024 – May 2025

ML and Data Pipeline Contributor

- Co-developed a haptic-glove teleoperation workflow with the simulation and control team to collect, QA, and version demonstrations for cube-stacking with a virtual arm and dexterous hand in NVIDIA Omniverse and Isaac Sim.
- Owned the learning-data pipeline module covering sequence formatting, preprocessing, and augmentation; integrated a multi-layer Transformer policy with RL fine-tuning for post-imitation improvement.
- Partnered with teammates to standardize logging and evaluation, troubleshoot failures across sim-data-training, and document the workflow to support repeatable experiments and smooth onboarding.

Power Consumption Prediction and Sensor-Integrated ML Pipeline

Jan 2024 – Jun 2024

ML Researcher / Energy Analytics Contributor

- Collaborated with a research team on an energy-focused ML project by integrating motherboard power and thermal sensors and building automated data collection and preprocessing pipelines.
- Developed an RNN-based power prediction model achieving $\sim 95\%$ accuracy; contributed to the publication at ICCT-Pacific 2025.